

Project Report: AI-Powered Passport Photo App

This report provides a summary for the development of the AI-powered instant passport photo application, from initial concept to it being functional.

1. Project Overview

The goal of this project was to create a web application that simplifies the process of getting a passport photo. The main challenge was to automate the removal of a photo's background and replace it with a solid color, a common requirement for official documents. The solution involved building an end-to-end system that includes a data pipeline, a deep learning model for background removal, and a web interface for user interaction.

2. Technical Components

2.1. Data and Annotation

Problem: To train an AI model to recognize a person, it needs a lot of examples. I started with a small dataset and segmented the dataset of 50 images properly

Solution: I used a tool called CVAT to manually draw a "mask" around the person in each of the 50 images. This mask tells the AI exactly which pixels belong to the person.

Data Augmentation: To make the model more robust, I used a technique called data augmentation. Using the Augmentations library, I created 250 new images by slightly changing the original 50 (e.g., rotating them, changing brightness, flipping them horizontally).

2.2. Model and Training

Model Choice: I chose a popular architecture for this task called U-Net because it's excellent at image segmentation. I also used a lightweight model called MobileNetV3 as a base to make the final model smaller and faster.

Training: Using the PyTorch framework, I trained the model on the augmented dataset of 250 images. I used Dice Loss to measure how well the model was doing and the Adam optimizer to help it learn. The training process ran for 25 epochs until the model became accurate enough to identify people in new photos.

2.3. App Development

Backend: I built a backend server using Flask that acts as the "brain" of the app. This server's main job is to take a photo, load the trained AI model, remove the background, and add a new one. I also wrote the code to save the trained model as a file (.pth) and load it correctly when the server starts.

Frontend: The frontend is the part the user sees, and I built it using HTML, CSS, and JavaScript. This is where the user uploads their photo and clicks the buttons to change the background and download the final image. The JavaScript code communicates with the backend server, sending the photo to be processed and receiving the finished result.

3. Conclusion

The project was successful in creating a functional, AI-powered passport photo application. The training process resulted in a model capable of accurately segmenting a person from their background. By separating the AI logic into a backend server, the application is both efficient and scalable. The web interface provides a simple and effective way for users to create compliant passport photos without the need for expensive software or a professional photographer.